

Assignment no7

1. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
KD2_Pritam_99277>SELECT *
-> FROM orders
-> WHERE cnum = (
->     SELECT cnum
->     FROM customers
->     WHERE cname = 'Cisneros'
-> );
```

onum	amt	odate	cnum	snum
3001	18.69	1990-10-03	2008	1007
3006	1098.16	1990-10-03	2008	1007

2 rows in set (0.00 sec)

2. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
KD2_Pritam_99277>select o.amt,c.cname,c.rating from customers c
-> inner join orders o
-> on o.cnum= c.cnum
-> where amt > @avgamt;
```

amt	cname	rating
5160.45	Liu	200
9891.88	Clemens	100
4723.00	Clemens	100

3 rows in set (0.00 sec)

3. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```
KD2_Pritam_99277>SELECT s.snum, s.sname, SUM(o.amt) AS total_amount
-> FROM salespeople s
-> INNER JOIN orders o
-> ON o.snum = s.snum
-> GROUP BY s.snum, s.sname
-> HAVING SUM(o.amt) > (SELECT MAX(amt) FROM orders);
+-----+-----+-----+
| snum | sname | total_amount |
+-----+-----+-----+
| 1001 | Peel  |      15382.07 |
+-----+-----+-----+
1 row in set (0.00 sec)
```

4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).

```
KD2_Pritam_99277>select * from customers
-> where rating >= any (select rating from customers where snum in (select
snum from salespeople where sname='serres'));
+-----+-----+-----+-----+-----+
| cnum | cname   | city   | rating | snum |
+-----+-----+-----+-----+-----+
| 2002 | Giovanni | Rome   | 200    | 1003 |
| 2003 | Liu      | San Jose | 200    | 1002 |
| 2004 | Grass    | Berlin | 300    | 1002 |
| 2008 | Cisneros | San Jose | 300    | 1007 |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

5. Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
KD2_Pritam_99277>select * from salespeople s
-> where s.city != all(select c.city from customers c);
```

snum	sname	city	comm
1007	Rifkin	Barcelona	0.15
1003	Axelrod	New York	0.10

```
2 rows in set (0.00 sec)
```

6. Write a query that selects all orders for amounts greater than any for the customers in London.

```
KD2_Pritam_99277>select * from orders
-> where amt > any(select amt from orders where cnum in(select cnum from
customers where city='london'));
```

onum	amt	odate	cnum	snum
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3008	4723.00	1990-10-04	2006	1001
3011	9891.88	1990-10-04	2006	1001

```
6 rows in set (0.00 sec)
```

7. Extract all the orders of Motika.

```
KD2_Pritam_99277>select * from orders
-> where snum in (select snum from salespeople where sname='Motika');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

8. Find all the orders of salespeople servicing customers in London.

```
KD2_Pritam_99277>select * from orders
-> where cnum in(select cnum from customers where city='london');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

9. Find names and numbers of all salesperson who have more than one customer.

```
KD2_Pritam_99277>SELECT s.snum, s.sname
-> FROM salespeople s
-> WHERE (
->     SELECT COUNT(*)
->     FROM customers c
->     WHERE c.snum = s.snum
-> ) > 1;
+-----+-----+
| snum | sname |
+-----+-----+
| 1001 | Peel  |
| 1002 | Serres |
+-----+-----+
2 rows in set (0.00 sec)
```

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
KD2_Pritam_99277>SELECT s.snum, s.sname, s.city,  
->      (SELECT COUNT(*)  
->      FROM customers c  
->      WHERE c.snum = s.snum) AS customer_count  
-> FROM salespeople s  
-> WHERE (SELECT COUNT(*)  
->      FROM customers c  
->      WHERE c.snum = s.snum) > 1;  
  
+-----+-----+-----+-----+  
| snum | sname | city | customer_count |  
+-----+-----+-----+-----+  
| 1001 | Peel | London | 2 |  
| 1002 | Serres | San Jose | 2 |  
+-----+-----+-----+-----+  
2 rows in set (0.01 sec)
```

11. Select customers who have a greater rating than any other customer in Rome.

```
KD2_Pritam_99277>select * from customers where rating> (select min(rating) f  
rom customers where city='rome');  
  
+-----+-----+-----+-----+-----+  
| cnum | cname | city | rating | snum |  
+-----+-----+-----+-----+-----+  
| 2002 | Giovanni | Rome | 200 | 1003 |  
| 2003 | Liu | San Jose | 200 | 1002 |  
| 2004 | Grass | Berlin | 300 | 1002 |  
| 2008 | Cisneros | San Jose | 300 | 1007 |  
+-----+-----+-----+-----+-----+  
4 rows in set (0.00 sec)
```

12. Select all orders that had amounts that were greater than at least one of the orders from '1990-10-04'.

```
KD2_Pritam_99277>select * from orders WHERE amt > ANY(select amt from orders where  
odate='1990-10-04');  
  
+-----+-----+-----+-----+-----+  
| onum | amt | odate | cnum | snum |  
+-----+-----+-----+-----+-----+  
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |  
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |  
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |  
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |  
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |  
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |  
| 3010 | 309.95 | 1990-10-04 | 2004 | 1002 |  
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |  
+-----+-----+-----+-----+-----+  
8 rows in set (0.01 sec)
```

13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
KD2_Pritam_99277>SELECT *
-> FROM orders o
-> WHERE o.amt < ANY (
->     SELECT o2.amt
->     FROM orders o2
->     INNER JOIN customers c
->     ON o2.cnum = c.cnum
->     WHERE c.city = 'San Jose'
-> );
```

onum	amt	odate	cnum	snum
3001	18.69	1990-10-03	2008	1007
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3007	75.75	1990-10-04	2004	1002
3008	4723.00	1990-10-04	2006	1001
3010	309.95	1990-10-04	2004	1002

8 rows in set (0.01 sec)

14. Select those customers whose rating are higher than every customer in London.

```
KD2_Pritam_99277>select * from customers where rating> all(select rating from customers where city='london');
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.00 sec)